

FAKE NEWS DETECTION SYSTEM FOR SOCIAL NETWORKS

A Project Report

submitted by

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for the award of the dual degree of*

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Place: Chennai

Date: 01.03.2026

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Shrinidhi Ramesh

ABSTRACT

The rapid spread of fake news on social media platforms presents a major challenge that requires detection systems capable of going beyond simply analyzing the text of an article. Traditional text-based methods often struggle because modern misinformation is intentionally designed to mimic credible news. To address this, this project presents a two-phase news detection pipeline that combines the analytical capabilities of Large Language Models (LLMs) with the structural mapping of Graph Neural Networks (GNNs).

This hybrid approach comprehensively evaluates both the written content itself and the social context of how that news spreads online.

Phase 1 focuses on deep content analysis and active verification. It uses an LLM to extract important features directly from news articles, generating metadata such as stance, sentiment, subjectivity, and clickbait indicators. Crucially, this phase also incorporates real-time online fact-checking using search agents like Tavily to retrieve external, verified context. An "LLM Judge" then evaluates these extracted features alongside the fact-checked evidence to provide a synthesized assessment of the article's credibility.

To overcome the limitations of relying solely on linguistic features, Phase 2 integrates this extracted data into a heterogeneous social graph. By modeling the distinct interactions between users and news articles—such as user-follows-user relationships, and who tweeted or retweeted specific content—the GNN analyzes the specific patterns of how information propagates through the network.

Ultimately, by combining what the news says, how it stands up to factual scrutiny, and how it travels, this pipeline creates a more reliable and resilient system for identifying fake news and exposing coordinated disinformation campaigns.

KEYWORDS: Fake News Detection, Large Language Models (LLM)
Graph Neural Networks (GNN), Automated Fact-Checking, LLM-as-a-Judge,
Heterogeneous Graphs, Information Propagation, Social Network Analysis

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ABBREVIATIONS

BERT	Bidirectional Encoder Representations from Transformers
GNN	Graph Neural Network
HetGNN	Heterogeneous Graph Neural Network
LLM	Large Language Model
CoT	Chain of Thought
R-GCN	Relational Graph Convolutional Network
GAT	Graph Attention Network
FRE	Flesch Reading Ease

CHAPTER 1

INTRODUCTION

1.1 Background

The proliferation of digital social networks has fundamentally transformed how news is generated, distributed, and consumed. While this democratization of information has enhanced global connectivity, it has simultaneously lowered the barrier to entry for publishing unverified, biased, or entirely fabricated content. "Fake news"—broadly categorized into misinformation (false information shared without malicious intent) and disinformation (deliberately crafted falsehoods)—spreads rapidly across social platforms, often outpacing the truth due to its sensationalized nature. The viral spread of such content poses significant threats to public trust, democratic processes, and social stability.

1.2 Problem Statement

The sheer volume and velocity of information generated on social media platforms make traditional, manual fact-checking by human investigative journalists practically impossible to scale. Existing automated detection systems often rely on shallow linguistic features or static blacklists, which fail to capture the nuanced, evolving nature of modern disinformation.

Specifically, current systems struggle with:

- **Contextual Dissonance:** Clickbait articles where the headline wildly exaggerates or contradicts the actual body text.
- **Subjective Masking:** Opinion pieces heavily masked as objective reporting using hedging and biased language.
- **Dynamic Claim Verification:** Static machine learning models cannot verify breaking news claims against real-time, real-world evidence.

Therefore, there is a critical need for an automated, scalable, and highly accurate system capable of understanding deep semantic context, cross-referencing claims dynamically, and analysing the social infrastructure through which the news spreads.

1.3 Motivation

The motivation for this project stems from the necessity to build a transparent and explainable AI system for media verification. Standard binary classifiers (tagging an article simply as "Fake" or "Real") offer no insight into why a decision was made. By leveraging the advanced reasoning capabilities of Large Language Models (LLMs) combined with real-world agentic search workflows, we can mimic the exact cognitive steps of a human fact-checker. This approach not only provides a high-confidence verdict but also produces an explainable trail of evidence—highlighting specific linguistic manipulations, stance mismatches, and contradictory external sources.

1.4 Scope of the project

To comprehensively address the complexities of online misinformation, this project is architected in two distinct phases, combining content-centric and network-centric approaches:

Phase 1: Content and Claim Verification (Focus of Chapter 1):

The development of a multi-agent LLM pipeline that ingests news articles, analyses linguistic markers (sentiment, stance, subjectivity, clickbait), and utilizes agentic web search to verify extracted claims against independent, high-authority domains.

Phase 2: Propagation and Social Context Analysis (To be discussed in Chapter 2):

The integration of Graph Neural Networks (GNNs) to map and analyse how these articles propagate through social networks, identifying fake users and fake news articles. Also see if the GNN can be used as a recommender to recommend news articles based on the user's preferences. [1]

1.5 About the Phase 1 System

The proposed solution for Phase 1 operates as an end-to-end automated pipeline that extracts, processes, and evaluates the veracity of online news articles. By leveraging a combination of fine-tuned Transformer models

(BERT) and Large Language Models (LLaMA 3.1 via Groq), the architecture

systematically analyzes linguistic features, structural dissonance, and domain authority.

Furthermore, the system employs an agentic search mechanism to cross-reference extracted claims against live web data, mimicking the workflow of a human investigative fact-checker.

CHAPTER 2

SOLUTION DETAILS

The solution is divided into two distinct but complementary phases: a content-centric pipeline (Phase 1) and a network-centric propagation analyzer (Phase 2)

2.1 Phase 1: Content-Centric AI Pipeline

The initial phase of the project focuses on developing the core content-analysis pipeline, which forms the foundation of the proposed fake news detection framework. Instead of relying on a single monolithic model, Phase 1 utilizes a multi-agent, modular architecture. The system integrates multiple linguistic, semantic, and credibility-oriented cues to holistically assess the authenticity of a news article before passing the aggregated data to an LLM for a final verdict.

2.1.1 System Architecture Overview

The workflow of the Phase 1 pipeline is designed to mimic the cognitive steps of a human fact-checker, automated through advanced NLP techniques:

- **Feature Extraction & Linguistic Analysis:** When an article is ingested, it is segmented into semantic chunks and independently analysed for various linguistic markers. This includes evaluating the emotional tone using **SieBERT** embeddings, detecting sensationalist clickbait phrasing in the headline, and computing readability metrics (such as the Flesch–Kincaid index) to flag unnatural linguistic complexity. Furthermore, the system identifies subjectivity, hedging, and bias markers—such as modal verbs and exaggerative qualifiers—that often correlate with manipulative writing.
- **Contextual Alignment & Entity Tracking:** A dedicated LLM module performs stance detection, evaluating whether the body text actually supports, refutes, or remains neutral toward the claims made in the headline. The system also extracts named entities (persons, organizations, locations) to analyse co-occurrence patterns and detect internal contradictions.

- **External Verification:** Moving beyond internal text analysis, the pipeline extracts factual claims and performs stance verification. It utilizes domain-based reputation scoring to evaluate the publication's source credibility and actively retrieves evidence from trusted repositories.
- **LLM-Based Judgment:** In the final stage, all extracted features and intermediate outputs—sentiment, stance, bias, readability, source credibility, and claim verification—are aggregated. This data packet is fed into a master Large Language Model acting as a meta-judge. The LLM performs holistic reasoning across these diverse features to assign the final veracity label (e.g., True, Misleading, Fake) along with a highly interpretable justification for its decision.

Collectively, these components enable a multi-perspective evaluation of news content, forming a robust base for the subsequent extensions in Phase 2.

Misinformation Detection Pipeline

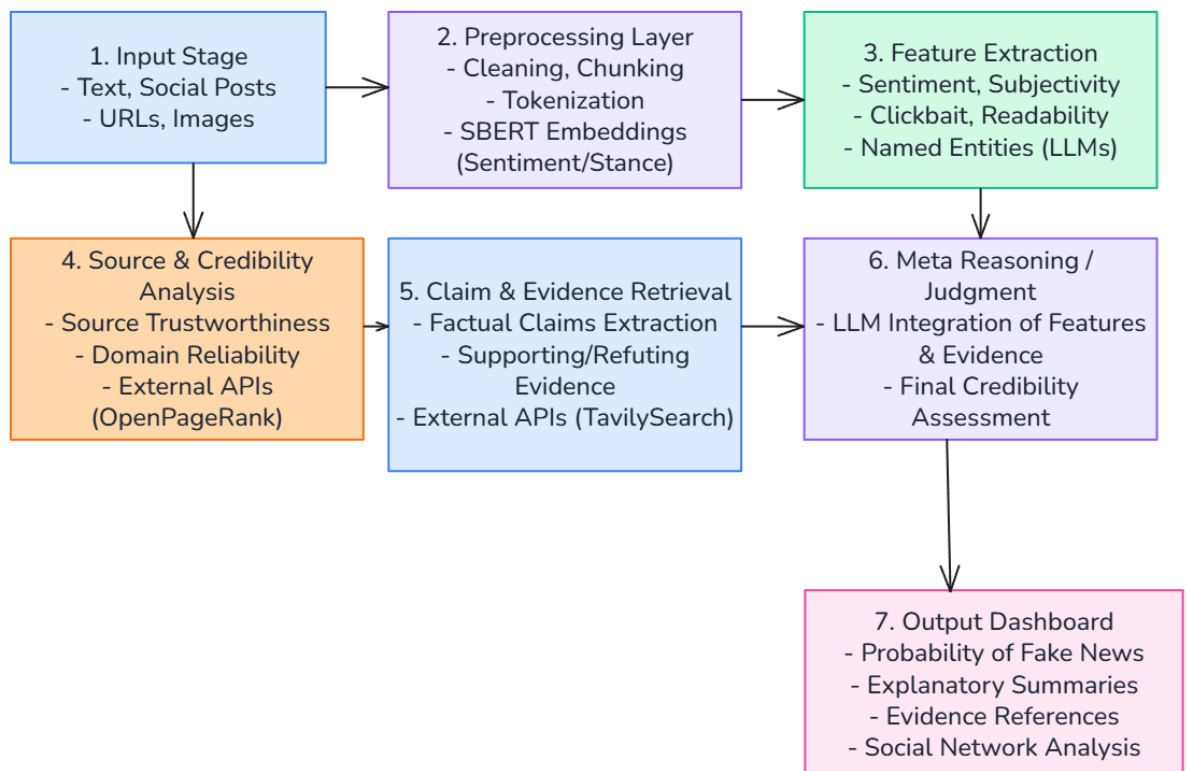


Figure 1 Phase 1 architecture diagram

2.2 Phase 2: Graph-Based Social Propagation Analysis

While Phase 1 determines the factual accuracy of a localized piece of text, fake news is uniquely characterized by its distribution mechanics. Misinformation often utilizes artificial amplification, coordinated bot networks, and rapid cross-platform sharing to gain traction. Phase 2 transitions from analysing isolated content, to analysing the complex network topology through which the content propagates, utilizing Graph Neural Networks (GNNs) [2].

2.2.1 Heterogeneous Graph Construction

To accurately model the social media environment, the system constructs a heterogeneous graph. Unlike homogeneous graphs where all nodes are identical, a heterogeneous graph allows the model to distinguish between different entities and the complex relationships between them. The proposed graph architecture consists of two primary node types: User Nodes and News Nodes. The edges connecting these nodes represent specific interactions, such as a user posting, sharing, or engaging with a specific news article.[3]

2.2.2 Node Feature Engineering and Integration

The success of a GNN relies heavily on the quality of the initial node features. This phase seamlessly integrates the outputs of Phase 1 to enrich the network analysis:

- **User Node Features:** User nodes are initialized with rich metadata extracted from social profiles. This includes demographic and behavioural data such as the user's location, profile name, biodata, account age, and historical engagement metrics.

- **News Node Features:** News nodes act as the critical bridge between the two phases.

Instead of relying solely on raw text, the news nodes are initialized using deep BERT embeddings to capture semantic meaning. Crucially, these embeddings are concatenated with the structured LLM-based features extracted during Phase 1—including the calculated stance dissonance, subjectivity score, hedging presence, sentiment and clickbait score.

2.2.3 GNN Architectures for Multi-Relational Data

Because social networks involve multiple types of relationships and data structures, standard convolutional networks are insufficient. The system will experiment with and benchmark advanced GNN architectures [3] tailored for complex topologies:

- **Relational Graph Convolutional Networks (R-GCN):** R-GCNs are specifically designed to handle highly multi-relational data within heterogeneous graphs. They apply relation-specific transformations, allowing the model to learn different weights for a "User-Shared-News" edge versus a "User-Follows-User" edge.
- **Graph Attention Networks (GAT):** GATs utilize masked self-attention layers to weigh the influence of neighbouring nodes differently. In the context of fake news, a GAT can learn to assign higher attention to highly influential nodes (e.g., super-spreaders or verified fact-checkers) within a propagation cascade, rather than treating all shares equally.

2.2.4 Core Predictive Tasks and Recommendations

The finalized graph architecture is optimized to perform three simultaneous downstream tasks, transforming the pipeline into a comprehensive security and analysis tool:

1. **Node Classification (News Veracity):** By evaluating both the rich semantic features of the News node (from Phase 1) and its structural position within the graph (from Phase 2), the GNN performs binary or multi-class node classification to definitively label the news as Fake or Real.
2. **Node Classification (User Authenticity):** The network evaluates the behavioral topology of User nodes to classify them as legitimate human users or automated malicious actors (bots). Users exhibiting dense, highly coordinated sharing patterns of low-veracity news are mathematically isolated.
3. **Link Prediction as a Recommender System:** Beyond classification, the GNN is adapted to function as a recommender system. By performing link prediction, the system can forecast which users are most susceptible to engaging with specific fabricated news clusters. Conversely, this same mechanism can be utilized to proactively recommend highly verified, corrective news articles to vulnerable network segments.

2.3 Chapter Summary and Organization of Subsequent Sections

Having established the high-level architecture and theoretical foundations of both the content-centric pipeline (Phase 1) and the graph-based propagation model (Phase 2), the following chapters will systematically deconstruct Phase 1.

The subsequent sections provide a granular, step-by-step technical breakdown of the individual modules comprising the core AI pipeline. Each module—ranging from sentiment and stance analysis to clickbait detection and agentic source verification—will be detailed individually to illustrate the feature extraction and reasoning mechanisms that drive the final veracity judgment.

CHAPTER 3

SENTIMENT AND STANCE ANALYSIS

The first analytical phase focuses on quantifying the emotional tone and structural consistency of the ingested news article. Misinformation relies heavily on emotional manipulation and sensationalist headlines that often contradict the actual body text.

3.1 Sentiment Analysis

To evaluate the emotional trajectory of an article, the system utilizes a chunk-based Transformer pipeline. Since full-length articles exceed standard model token limits, a recursive processing architecture is implemented:

- **Recursive Text Chunking:** The article is parsed using a `RecursiveCharacterTextSplitter`, which divides the text into overlapping segments (400 characters with a 50-character overlap) to preserve semantic context across chunk boundaries.
- **Chunk-Level Scoring:** Each segment is independently processed through a pre-trained HuggingFace sentiment-analysis pipeline, generating a specific sentiment label and confidence score per chunk.
- **Majority Voting Aggregation:** The system aggregates these chunk-level predictions using a weighted majority voting algorithm. The scores are summed and normalized, and the label with the highest aggregate score is assigned as the final document-level sentiment.

3.2 Stance Analysis

Fake news frequently employs contextual dissonance, where an inflammatory headline is not supported by the underlying text. The stance module mathematically quantifies this discrepancy using a Large Language Model.[4]

- **Structured LLM Inference:** The system isolates the headline and body_text and processes them through LLaMA 3.1. To ensure deterministic outputs, the model is constrained by a strict Pydantic schema (Stance Output).
- **Scoring and Explainability:** The LLM evaluates the contextual relationship and returns three specific data points:
 1. **Stance Label:** Categorizes the body text as "Favor," "Against," or "Neutral" relative to the headline.
 2. **Dissonance Score:** A continuous value from 0.0 to 1.0, where higher scores indicate a stronger contradiction between the headline and the body.
 3. **Reasoning:** A chain-of-thought string providing an explainable justification for the stance classification.

3.2.1 Prompt Design for Stance Detection

To ensure the LLM strictly evaluates the contextual dissonance without hallucinating, the following prompt template is utilized:

```
head_body_stance_template = """
<|user|>
You are a forensic linguistic auditor. Your task is to determine if an article's BODY
supports or contradicts its HEADLINE.

### INPUT DATA:
**HEADLINE:** "{headline}"
**BODY TEXT:** "{body_text}"

### STANCE DEFINITIONS:
- **FAVOR (Agree):** The body text provides facts and context that directly confirm the
headline's claim.
- **AGAINST (Disagree):** The body text explicitly contradicts or debunks the claim
made in the headline.
- **NEUTRAL (Unrelated/Discuss):** The body text mentions the same topic but does not
take a position, or it discusses a different aspect entirely (making the headline "unrelated").

### EVALUATION LOGIC:
1. Identify the core claim of the Headline.
2. Check if the Body contains evidence for that specific claim.
3. If the Body is about a different topic, label as NEUTRAL/UNRELATED.

### OUTPUT FORMAT:
Provide your output EXCLUSIVELY in a JSON object:
{{
  "stance_label": "Favor/Against/Neutral",
  "dissonance_score": 0.0 to 1.0 (Higher means the body is UNRELATED or
CONTRADICTS the head),
  "reasoning": "Identify the specific claim in the head and whether the body addresses it."
}}
do not add any explanations outside the JSON.
do not return any text other than the JSON object.
<|end|>
<|assistant|>
"""
```

CHAPTER 4

SUBJECTIVITY AND HEDGING / BIAS MARKERS

4.1 Subjectivity Markers

4.1.1 Identifying Subjectivity

To distinguish between factual reporting and editorializing, the pipeline utilizes a Chain-of-Thought (Cot) prompting strategy [6]. The LLM acts as a linguistic analyst, categorizing the text strictly as "Objective" or "Subjective" based on universally accepted journalistic principles.

- **Objective Assessment:** The model scans for "Anchors" (Named Entities such as People, Places, Dates) and verifiable "Speech Acts" (e.g., "Official records show").
- **Subjective Assessment:** The model evaluates the text for first-person bias, unverified speculation, and "value-laden" unattributed adjectives.
- **Structured Output:** The system extracts the identified entities, a continuous confidence score (0.0 to 1.0), and a chain-of-thought reasoning string that explicitly justifies the classification.

4.1.2 Prompt Design for Subjectivity Analysis

The following prompt template constrains the LLM to output a strict JSON evaluation based on the defined criteria:

<|user|>

You are a professional linguistic analyst. Your task is to classify the provided text 'S' as 'Objective' or 'Subjective' based on universal journalistic principles.

CLASSIFICATION CRITERIA:

*1. **OBJECTIVE (Hard News/Reporting):***

*- **Anchors:** Uses Named Entities (People, Places, Organizations, Dates) as the primary subjects.*

*- **Attribution:** Uses "Speech Acts" to relay information (e.g., "X said", "According to Y", "Official records show").*

*- **Nature:** Describes verifiable actions, events, or official statements. Even if the event is intense or tragic, the language remains reportorial and descriptive.*

*2. **SUBJECTIVE (Editorial/Opinion):***

*- **First-Person/Direct Bias:** Uses the author's voice (e.g., "I believe", "We must", "It is my view").*

*- **Unattributed Adjectives:** Uses "value-laden" words without quoting a source (e.g., "The disgusting behavior", "A magnificent achievement", "This shameful act").*

*- **Speculation:** Makes claims about people's internal thoughts, feelings, or future intentions that cannot be verified.*

STEP-BY-STEP ANALYSIS:

*1. **Identify Entities:** List the people, organizations, or locations mentioned.*

*2. **Scan for Attribution:** Identify if the claims are linked to a specific source or if they are "naked" claims by the author.*

*3. **Tone Evaluation:** Check if adjectives are used to describe a person's character (Subjective) or a person's verifiable action (Objective).*

S: {text}

Provide your output EXCLUSIVELY in a JSON object:

{{

"entities_identified": ["Entity1", "Entity2"],

"final_category": "Objective/Subjective",

"score": 0.0 to 1.0,

"reasoning": "Explain the decision by identifying if the text relies on named entities and attributed statements (Objective) or author-driven opinion/adjectives (Subjective)."

}}

do not add any explanations outside the JSON.

do not return any text other than the JSON object.

<|end|>

<|assistant|>

4.2 Hedging / Bias Markers

Misinformation often employs epistemic modality—words that soften a claim or express vagueness (e.g., "might," "could suggest," "reportedly")—to avoid definitive, verifiable statements. The Hedging Detection module isolates this specific linguistic vulnerability.

- **Step-by-Step Scanning:** The LLM scans the text specifically for expressions of uncertainty or extreme caution.
- **Impact Evaluation:** It calculates how these softening phrases degrade the overall certainty and reliability of the stated facts.
- **Structured Output:** The module returns a Boolean flag (`hedging_detected`), a normalized score (0.0 representing absolute certainty, 1.0 representing high vagueness), and extracts the exact hedge words utilized by the author.

4.2.1 Prompt Design for Hedging Detection

To ensure the LLM strictly isolates uncertainty markers without conversational padding, the following prompt is utilized:

You are an expert in linguistic analysis. Your task is to determine if the text 'S' contains hedging language (expressions of uncertainty, caution, or vagueness).

Think step-by-step:

- 1. First, scan the text for words or phrases that soften a claim, express uncertainty, or show caution (e.g., "seems", "possible", "could suggest", "probably").*
- 2. Next, analyze the impact of these phrases on the overall certainty of the text.*
- 3. Finally, based on the presence and strength of these hedging phrases, determine if hedging is present and assign a score.*

Return ONLY a valid JSON object.

Do NOT wrap in markdown.

Do NOT include any text before or after the JSON.

Do NOT include trailing commas.

CRITICAL: Return ONLY valid JSON. Do not use unescaped double quotes inside string values. If you need to quote something inside a text field, use single quotes (e.g., 'young' instead of "young").

The JSON must match EXACTLY this structure:

```
{{  
  "hedging_detected": A boolean value (true or false).  
  "hedging_score": A float between 0.0 (no hedging) and 1.0 (very high hedging).  
  "reasoning": A concise, step-by-step explanation of your analysis, identifying  
the specific hedge words found.  
}}  
do not add any explanations outside the JSON.  
do not return any text other than the JSON object.
```

S: {text}

CHAPTER 5

CLICKBAIT AND READABILITY ANALYSIS

Misinformation often relies on highly sensationalized headlines to drive engagement and employs anomalous writing complexities compared to professional journalism. This phase utilizes a combination of fine-tuned Transformer networks and classical NLP heuristics to quantify these traits.

5.1 Clickbait Analysis

5.1.1 Transformer-Based Detection

To detect sensationalism purely designed for engagement, the system isolates the article's headline and processes it through a fine-tuned BERT architecture (Stremie/bert-base-uncased-clickbait). Unlike rule-based keyword matchers, this transformer model analyzes the syntactic patterns and contextual framing of the headline. Initially the use of LLMs was considered but BERT models have proven to be more accurate at clickbait detection than LLMs. Hence the decision to use a fine-tuned BERT was considered.[5]

- **Execution Logic:** The headline is passed through the classification pipeline. If the model outputs LABEL_1, the text is flagged as clickbait.
- **Scoring Output:** The module calculates a normalized confidence score between 0.0 and 1.0. A dictionary is returned containing a Boolean flag (clickbait_detected) and the precise clickbait_score, which is later fed into the master LLM judge. Error handling is built-in to bypass analysis and return neutral scores (0.0) if the headline is empty or malformed.

5.2 Readability and Lexical Features

5.2.1 Classical NLP Readability Metrics

Rather than using deep learning, this module employs classical linguistic formulas via the textstat library to evaluate the structural complexity of the body text.

- **Metric Calculation:** The system calculates two standard metrics:
 1. **Flesch Reading Ease (FRE):** Evaluates sentence length and syllable count to determine how difficult the text is to understand.
 2. **Flesch-Kincaid Grade Level:** Translates the FRE score into an approximate educational grade level required to comprehend the text.
- **Categorization:** Based on the Flesch Reading Ease score, the system categorizes the text into strict readability tiers:
 - **Easy:** Score > 70
 - **Moderate:** Score > 30 and < 70
 - **Hard:** Score < 30
- **Safety Thresholds:** To ensure statistical validity, the module enforces a minimum character limit (50 characters), returning neutral or unknown values for texts that are too short to produce reliable readability metrics.

CHAPTER 6

SOURCE CREDIBILITY HEURISTICS

The final analytical phase before synthesizing the final verdict moves beyond internal linguistic analysis to external verification. This module evaluates the historical authority of the publishing domain and actively cross-references the article's core claims against real-time web evidence.[7]

6.1 Domain Authority (PageRank)

Before analysing the content, the system evaluates the hosting infrastructure. Fake news is frequently published on newly registered, low-authority, or spoofed domains.

API Integration: The system extracts the domain from the target URL and queries the OpenPageRank API.

Authority Scoring: This returns a global PageRank integer. A high score indicates a historically trusted, well-linked domain, whereas a score of 0 or near zero flags the source as potentially untrustworthy or newly established.

6.2 Agentic Fact-Checking and Evidence Verification

This module acts as an autonomous AI agent, mimicking the workflow of a human investigative fact-checker to verify the claims made in the article.[8]

1. Claim Extraction and Search:

The system utilizes an LLM to distil the article down to a single, dense paragraph of factual claims (preserving named entities and dates).

It extracts keywords from this claim and triggers the Tavily Search API to find the latest independent news reports covering the same topic.

2. Evidence Scraping:

The agent scrapes the content from the top search results, explicitly skipping any URL that matches the original target article to prevent self-referencing. The scraped evidence is trimmed to fit within token limits.

3. LLM Comparative Analysis:

The original claim and the scraped independent evidence are fed into an LLM fact-checking prompt.

The LLM performs a step-by-step comparison, analysing whether the main actors, actions, locations, timelines, and numbers match.

4. Scoring and Verdict Aggregation:

For each independent source, the LLM assigns a verdict: "Supported", "Refuted", "Mixed", or "Irrelevant", alongside a confidence score.

The system aggregates these scores using a weighted mapping logic. If the total confidence heavily favours "Supported" or "Refuted," that becomes the final evidence verdict. If there is a close split (less than 20% difference) between supporting and refuting evidence, the system intelligently labels the claim as "Disputed."

The module returns this final verdict, the aggregate confidence score, and specific evidence quotes extracted from the external sources.

6.2.1 Prompt Design for Claim Verification

The following prompt is used for the claim verification step:

You are a senior investigative fact-checker.

Your task is to verify the truthfulness of the ARTICLE CLAIM using multiple independent EVIDENCE SOURCES.

Treat the evidence sources as reliable factual reporting.

=====

ARTICLE CLAIM:
{article_claim}

=====

Below are independent evidence sources labeled as SOURCE 1, SOURCE 2, etc.

{combined_evidence}

=====

ANALYSIS INSTRUCTIONS

=====

For EACH source separately:

1. Identify key factual elements in the ARTICLE CLAIM:

- *Main actors*
- *Actions*
- *Location*
- *Time references*
- *Numbers or scale*
- *Stated motive or cause (if any)*

2. Identify the key factual elements present in the SOURCE.

3. Compare both carefully:

- *Are the same actors involved?*
- *Are the actions described the same?*
- *Do locations and timelines match?*
- *Are numbers consistent?*
- *Is any crucial context missing or altered?*

4. Determine verdict for that source:

- *"Supported"*
- *"Refuted"*
- *"Mixed"*
- *"Irrelevant"*

A claim is:

- *"Supported" if the source clearly confirms the claim's core facts.*
- *"Refuted" if the source clearly contradicts the core facts.*
- *"Mixed" if partially correct but misleading, exaggerated, outdated, or missing some of the context.*
- *"Irrelevant" if the source does not meaningfully address the claim.*

=====

OUTPUT FORMAT

=====

Return a JSON object with a single key "results" that is a list of entries.

Each entry must follow this structure:

- *"source_id": integer (1, 2, 3...)*
- *"step_by_step_analysis": string (brief comparison of key entities, actions, numbers, timeline)*
- *"evidence_quote": string or null (exact quote from source; null if irrelevant)*
- *"verdict": one of "Supported", "Refuted", "Mixed", "Irrelevant"*
- *"confidence_score": float 0.0 to 1.0*

Do NOT add explanations outside the JSON.

""""

CHAPTER 7

LLM-BASED JUDGMENT

The previous modules in Phase 1 operate as isolated heuristic agents, extracting granular data regarding an article's emotional tone, structural dissonance, linguistic bias, and external factual accuracy. However, no single metric can definitively classify a complex piece of misinformation. The final stage of the Phase 1 pipeline introduces a master LLM judge to synthesize these disparate signals into a cohesive, explainable verdict.[8]

7.1 Final Verification

To mimic the holistic reasoning of a human editor, the system compiles all intermediate outputs into a comprehensive data packet.

The judge- input packet aggregates the following metrics:

Linguistic Signals: Sentiment scores, subjectivity type and score, hedging intensity, clickbait probability, and readability complexity.

Contextual Signals: Stance label and dissonance score (evaluating headline vs. body text).

Credibility & Verification Signals: PageRank score, agentic fact-check verdict, aggregate verification confidence, and specific evidence quotes.

The Master Judge Execution: This aggregated packet is passed to a Large Language Model (LLaMA 3.1) configured with a specialized master-judge-template.

The Master Judge evaluates the interplay between the metrics—for instance, recognizing that an article with high subjectivity, strong clickbait markers, low PageRank, and a "Refuted" fact-check verdict is highly likely to be fabricated.

7.2 Structured Veracity Output

The LLM judge outputs a structured JSON object containing four critical evaluations:

Veracity Label: The definitive classification of the article, restricted to four distinct categories: *Real*, *Fake*, *Probable Real*, or *Probable Fake*.

Reliability Score: A continuous floating-point value (0.0 to 1.0) representing the system's overall confidence in the assigned veracity label.

Data Quality Weight: A metric (0.0 to 1.0) evaluating how useful this specific article and its analysis would be as training data for future models, fulfilling the objective of

autonomous continuous learning.

Justification: A concise, explainable summary (maximum of six sentences). This reasoning string explicitly references the PageRank, the agentic fact-check results, and the linguistic signals to transparently explain why the article received its final label. By utilizing this LLM-based judgment, the Phase 1 pipeline ensures that the final fake news classification is not a black-box output, but rather a deeply reasoned and explainable conclusion.

7.2.1 Prompt Design for LLM Based Judgment

The following prompt template bounds the LLM's reasoning to the specific heuristic signals extracted during Phase 1:

```
Master_judge_template = """
You are the Final Veracity Arbiter. Your task is to determine if a news article is Real, Fake, or
somewhere in between based on the following specific data signals.

### INPUT DATA SIGNALS:
1. Sentiment Analysis: {sentiment_scores}
2. Subjectivity Type: {subjectivity_type}
3. Subjectivity Score: {subjectivity_score}
4. Hedging Detected: {hedging_detected}
5. Hedging Score: {hedging_score}
6. Clickbait Detected: {clickbait_detected}
7. Clickbait Score: {clickbait_score}
8. Readability Score: {readability_score}
9. Readability Complexity: {readability_complexity}
10. Source Authority (PageRank): {pagerank_score}
11. Fact-Check Verification Verdict: {verdict}
12. Fact-Check Confidence: {verify_conf}
13. Fact-Check Evidence Quote: {evidence_quote}
14. Stance Analysis label: {stance_label}
15. Stance Dissonance Score: {dissonance_score}

### JUDGMENT CATEGORIES & LOGIC:
```


- ****Real:****
 - * *Fact check verdict* : if the fact check verdict is "Supported" with a high confidence score (e.g., >0.8), this is the strongest signal of authenticity and should heavily influence the final label.
 - * *Tone*: High Objectivity.
 - * *Language*: Low or no Hedging; low Clickbait.
 - * *Readability*: Balanced (not extremely simple/childish nor overly convoluted).
 - * *Authority*: High PageRank combined with a "Supported" Fact-Check Verdict.
 - * *Stance* : Article body supports the headline claim (stance_label : Favor) with low dissonance score.
- ****Probable Real:****
 - * *Fact-Check Verdict* is "Supported" but PageRank is low (e.g., local news).
 - * *OR PageRank* is high but *Fact-Check* is "Mixed" (minor contextual errors in a reliable source).
 - * *Stance* : Article body mostly supports the headline claim (stance_label : Favor) with moderate dissonance score.
- ****Probable Fake:****
 - * *Fact-Check* is "Mixed" or "Refuted" + High Clickbait
 - * *Language*: Highly Subjective and heavy use of Hedging (avoiding direct responsibility for claims).
 - * *Stance* : Article body contradicts or is unrelated to the headline claim (stance_label : Against/Neutral) with high dissonance score.
- ****Fake:****
 - * *Fact-Check Verdict* is "Refuted" + Low Source Authority (PageRank).
 - * *Tone*: Heavily Biased/Subjective and high Clickbait score.
 - * *Note*: Even if PageRank is high, a "Refuted" Fact-Check Verdict for a specific claim can override source authority.
 - * *Stance* : Article body contradicts the headline claim (stance_label : Against) with very high dissonance score.

OUTPUT FORMAT (Strict JSON):

```
{
  "veracity_label": "Real/Fake/Probable Real/Probable Fake",
  "reliability_score": 0.0 to 1.0 (How much should we trust this specific article?),
  "data_quality_weight": 0.0 to 1.0 (How clean/useful is this data point for training a model?),
  "justification": "Provide a concise summary (max 6 sentences) explaining how you balanced the 9 signals. Specifically reference how PageRank, Fact-Check, and linguistic style (Subjectivity/Readability) influenced the final label."
}
```

Do not add any explanations outside the JSON.

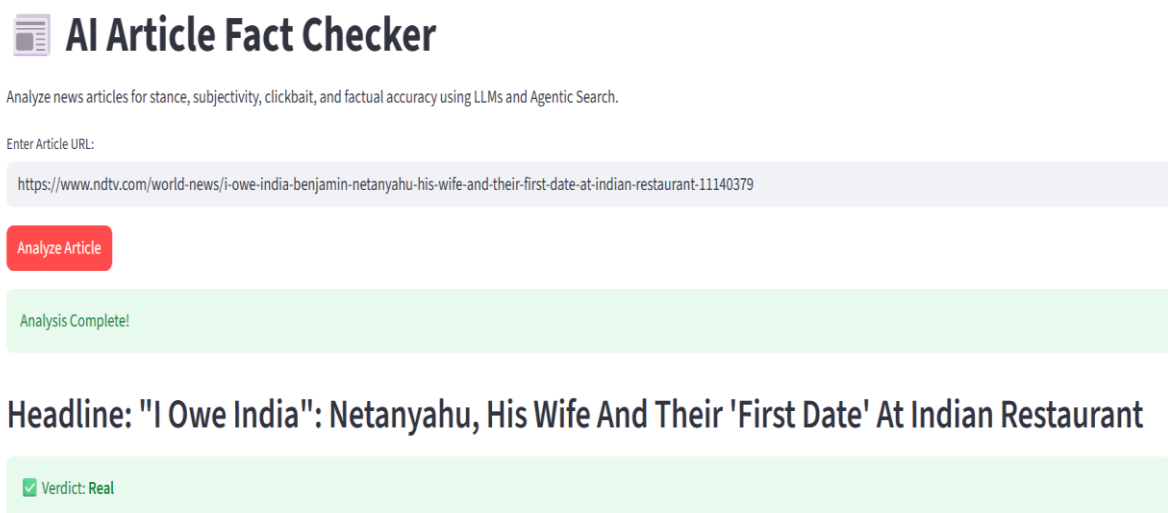
Do not return any text other than the JSON object.

""

CHAPTER 8

INTERACTIVE DASHBOARD

For the Phase 1 implementation, an interactive Streamlit dashboard was developed to provide a transparent, real-time interface for the classification pipeline. Rather than acting as a general visualization tool, this dashboard serves as a diagnostic window into the model's assessment of an article's authenticity. By integrating a veracity meter and displaying individual signal scores, the application allows for a granular analysis of why a specific article is flagged as real or fake.



AI Article Fact Checker

Analyze news articles for stance, subjectivity, clickbait, and factual accuracy using LLMs and Agentic Search.

Enter Article URL:

<https://www.ndtv.com/world-news/i-owe-india-benjamin-netanyahu-his-wife-and-their-first-date-at-indian-restaurant-11140379>

Analyze Article

Analysis Complete!

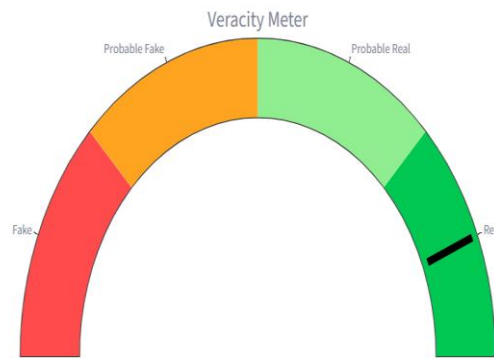
Headline: "I Owe India": Netanyahu, His Wife And Their 'First Date' At Indian Restaurant

✓ Verdict: Real

Figure 2 The URL has to be entered here

Headline: "I Owe India": Netanyahu, His Wife And Their 'First Date' At Indian Restaurant

✓ Verdict: Real



Justification: The article is labeled as Real due to the high confidence 'Supported' Fact-Check Verdict, high PageRank, and objective tone. The low hedging and clickbait scores, balanced readability, and high source authority further support this label. The stance analysis label of Favor with a low dissonance score also aligns with the Real category. The only minor concern is the moderate readability complexity, but it does not outweigh the strong signals of authenticity. Overall, the article's linguistic style and data quality are high, making it a reliable source of information. The reliability score is high due to the strong Fact-Check Verdict and high PageRank, while the data quality weight is moderate due to the moderate readability complexity.

Figure 3 The veracity meter and justification from the LLM

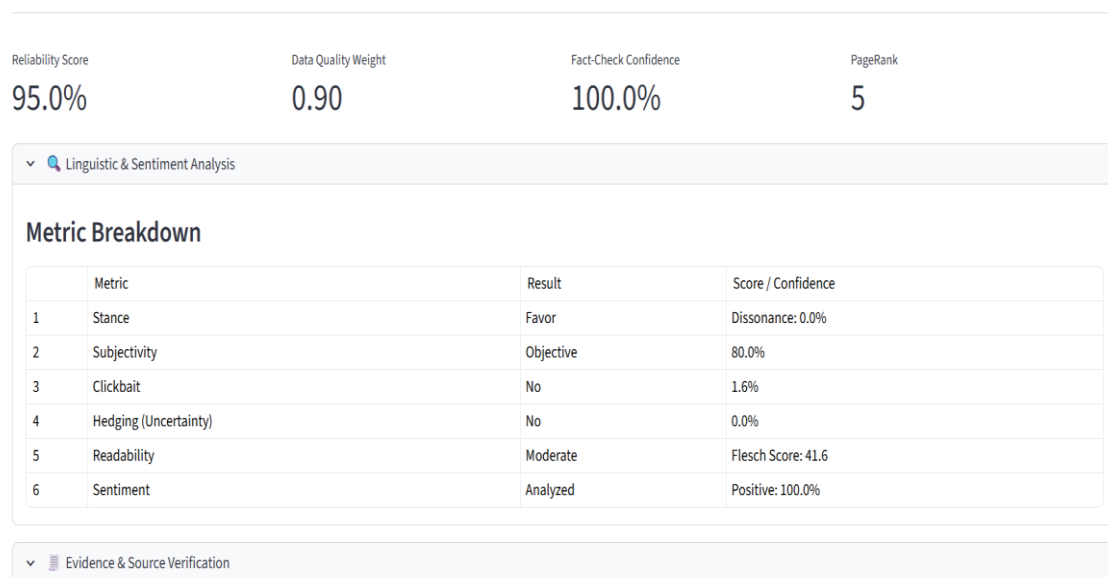


Figure 4 Linguistic analysis

Evidence & Source Verification

Final Evidence Verdict: Supported

Sources Checked: 1

3 - Verdict: Supported

Link: <https://economictimes.indiatimes.com/news/new-updates/i-owe-india-a-great-personal-debt-benjamin-netanyahu-recalls-first-date-at-indian-restaurant/articleshow/128816330.cms>

Analysis: Main actors: Netanyahu, Sara, Reena Pushkarna. Actions: Netanyahu met Sara at an Indian restaurant in Tel Aviv. Location: Tel Aviv. Time reference: No specific date mentioned. Numbers: None. Stated motive or cause: Netanyahu owes India a great personal debt.

Quote: "I owe India a great personal debt. When I met Sara for the first time, I think our first or second date was in an Indian restaurant in Tel Aviv. It was Reena Pushkarna."

Figure 5 Fact check verdict

CHAPTER 9

RESULTS AND ACCURACY ANALYSIS

The current pipeline was tested on 26 real and 27 fake news article URLs from the PolitiFact dataset from the Fake News Net Benchmark dataset repository.

The following results were obtained on the PolitiFact real dataset :

	precision	recall	f1-score	support
Fake	0.00	0.00	0.00	0
Real	1.00	0.85	0.92	26
accuracy			0.85	26
macro avg	0.50	0.42	0.46	26
weighted avg	1.00	0.85	0.92	26

Figure 6 Results for PolitiFact Real dataset

- **Accuracy (85%):** The pipeline correctly identified 85% of the total articles processed.
- **Precision (1.00):** When the pipeline predicted an article was "Real," it was correct 100% of the time (zero False Positives).
- **Recall (0.85):** The pipeline successfully identified 85% of all actual "Real" articles in the set, but missed 15% (False Negatives).

The following results were obtained on the PolitiFact Fake dataset :

	precision	recall	f1-score	support
Fake	1.00	0.52	0.68	27
Real	0.00	0.00	0.00	0
accuracy			0.52	27
macro avg	0.50	0.26	0.34	27
weighted avg	1.00	0.52	0.68	27

Figure 7 Results for PolitiFact Fake Dataset

- **Accuracy (52%):** The model struggled significantly more with this set, correctly identifying only 14 out of 27 "Fake" articles.
- **Precision (1.00):** Similar to the real set, when the model did commit to a "Fake" verdict, it was 100% accurate. It never called a "Real" article "Fake" in this specific batch.
- **Recall (0.52):** This is the weakest point. The model missed nearly half (48%) of the "Fake" articles, likely misclassifying them as "Real" or "Probable Real" because they "looked" legitimate.

CONCLUSION AND FUTURE SCOPE

Conclusion

The proposed system establishes a robust, two-phase framework for tackling the complex challenge of digital misinformation. By transitioning from a content-centric LLM analysis to a context-aware GNN approach, the pipeline moves beyond simple text classification. It successfully captures not just what is being said, but how it spreads and who is amplifying it. This hybrid architecture ensures that even sophisticated misinformation, which might bypass linguistic filters, is flagged by its abnormal propagation patterns.

Future Scope

Future enhancements to the pipeline will introduce :

- A dedicated image-fake detection step to extract visuals from news articles, analyzing them for manipulation and verifying their relevance to the content.
- Additionally, the system will evolve into an agentic architecture by wrapping individual feature-extraction steps—such as stance analysis and fact-checking—into modular, agentic nodes.
- **Improving Accuracy on Fake Datasets** : A primary focus will be enhancing the system's ability to detect misinformation (currently at **52% recall**) by integrating image veracity signals and refining the LLM prompts to be more "skeptical" of linguistic patterns common in fake news.
- Finally, Phase 2 will launch as an open-ended research effort exploring how GNNs can be enhanced by incorporating rich, LLM-derived feature embeddings and veracity labels as node metadata. This hybrid GNN-LLM approach will ultimately be used to map social propagation patterns and detect coordinated disinformation behavior networks.

APPENDIX

A.1 Section 1

A.1.1 Subsection heading

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A.2 Section 2

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Weekly Review Report

Image of the Weekly review report

FAKE NEWS DETECTION SYSTEM FOR SOCIAL NETWORKS

Misinformation and fake news have become critical threats in today's digital landscape. With social media shaping opinions and influencing decisions, the spread of false information can have serious consequences — from social unrest to public health crises.

This project aims to develop a comprehensive AI framework for fake news detection, combining Large Language Models (LLMs), graph-based social network analysis, and autonomous AI agents.

The goal is to create a scalable, explainable, and adaptive system that not only detects fake news but also understands its propagation dynamics.

OBJECTIVES:

- 1) Design a modular AI pipeline that combines text and social context for fake news detection.
- 2) Develop a hybrid reasoning system integrating LLMs and GNNs for multimodal misinformation analysis.

Plagiarism Report

